

MEMORA: A SMART FLASHCARDS LEARNING APPLICATION

ABSTRACT:

Memora is a smart flashcards learning application developed using Flutter and Firebase to provide an interactive and effective learning experience. The application enables users to create and manage flashcard decks, study concepts through flip-card interactions, and evaluate their understanding using quiz-based assessments. It also includes a progress tracking feature that allows users to monitor their learning activities and performance over time. Firebase Authentication ensures secure user access, while Cloud Firestore provides real-time data storage and synchronization.

Developed with Flutter, the application offers a responsive and user-friendly interface that supports multiple platforms. By combining active recall techniques with modern mobile technology, Memora helps learners improve knowledge retention, stay organized, and engage in self-paced learning. The system is designed to support students and lifelong learners by providing a simple, accessible, and efficient digital platform for enhancing academic performance and learning outcomes.

Keywords— Flutter, Firebase, Flashcards, Mobile Learning, Quiz System, Cloud Firestore, Progress Tracking.

INTRODUCTION:

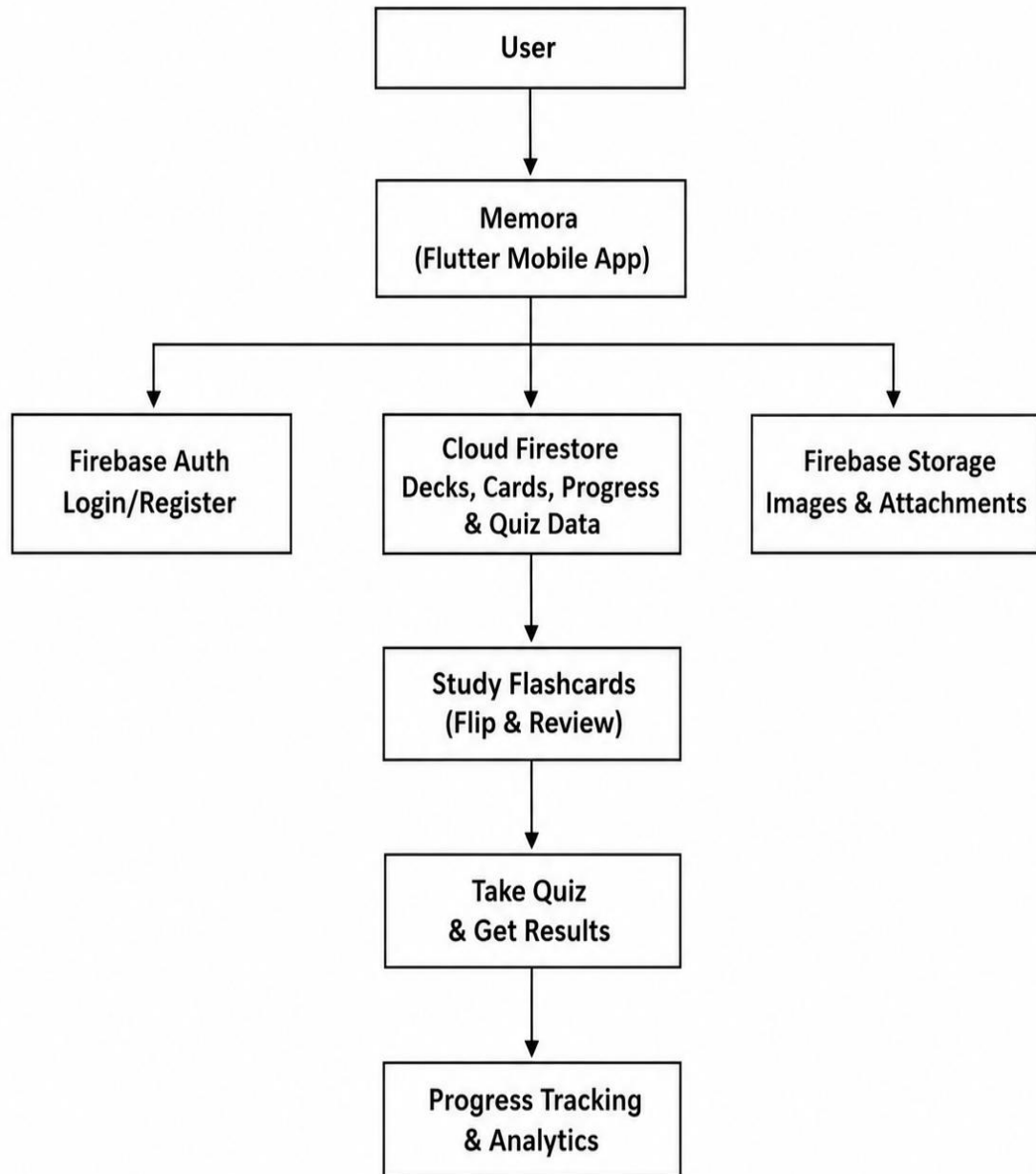
In today's digital era, students require learning tools that are interactive, accessible, and effective. Traditional study methods often make it difficult to organize information and track learning progress. Memora is a Smart Flashcards Learning Application developed using Flutter and Firebase to provide a modern and engaging learning experience. The application allows users to create flashcard decks, study using interactive flip cards, take quizzes, and monitor their performance through progress tracking. By combining active recall techniques with cloud-based technology, Memora helps learners improve knowledge retention, stay organized, and achieve their learning goals more efficiently.

RELATED WORK:

Several digital learning applications have been developed to improve knowledge retention through flashcard-based learning. Applications such as Anki, Quizlet, and Brainscape provide features like digital flashcards, quizzes, and study assistance tools. Anki uses a spaced repetition technique to help users remember information effectively, while Quizlet offers interactive learning modes and collaborative study features. Brainscape focuses on confidence-based repetition to enhance memory retention. Although these applications are effective, many of them have complex interfaces, limited customization options, or require premium subscriptions for advanced features. To address these limitations, Memora is designed as a simple, user-friendly, and cloud-based learning platform that combines flashcard creation, quiz assessments, progress tracking, and secure data synchronization within a single application. By integrating Flutter and Firebase technologies, Memora provides an accessible and efficient learning experience suitable for students and lifelong learners.

PROPOSED WORK:

The proposed system, Memora, is designed to provide an interactive and efficient learning platform that helps users improve knowledge retention through digital flashcards and self-assessment quizzes. The application allows users to create and manage flashcard decks, study using flip-card interactions, and evaluate their understanding through quiz-based learning. To ensure secure access and reliable data management, Firebase Authentication is used for user authentication and Cloud Firestore is used for real-time data storage and synchronization. The system also includes a progress tracking module that enables users to monitor their learning activities, quiz performance, and study achievements. Developed using Flutter, Memora offers a responsive and user-friendly interface that supports multiple platforms. By combining active recall learning techniques with modern mobile and cloud technologies, the proposed system provides a simple, accessible, and effective solution for students and lifelong learners to enhance their learning experience.



Important Code Snippets:

- **Main.dart**

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'package:firebase_core/firebase_core.dart';

import 'firebase_options.dart';
import 'providers/deck_provider.dart';
import 'screens/splash/splash_screen.dart';

Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized();

  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );

  runApp(
    MultiProvider(
      providers: [
        ChangeNotifierProvider(
          create: (_) => DeckProvider(),
        ),
      ],
      child: const MemoraApp(),
    ),
  );
}

class MemoraApp extends StatelessWidget {
  const MemoraApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Memora',
      theme: ThemeData(
        primarySwatch: Colors.indigo,
```

```

    ),
    home: const SplashScreen(),
  );
}
}

```

- **splash_screen.dart**

```

import 'dart:async';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import '../main/main_screen.dart';
import '../welcome/welcome_screen.dart';

class SplashScreen extends StatefulWidget {
  const SplashScreen({super.key});
  @override
  State<SplashScreen> createState() => _SplashScreenState();
}

class _SplashScreenState extends State<SplashScreen> {
  @override
  void initState() {
    super.initState();
    checkUser();
  }
  Future<void> checkUser() async {
    await Future.delayed(const Duration(seconds: 3));
    if (!mounted) return;
    final user = FirebaseAuth.instance.currentUser;
    if (user != null) {
      bool isAdmin = false;
      try {
        DocumentSnapshot userData = await FirebaseFirestore.instance
          .collection("users")
          .doc(user.uid)
          .get();
        if (userData.exists) {
          Map<String, dynamic> data = userData.data() as Map<String, dynamic>;
          isAdmin = data["role"] == "admin";

```

```

    }
  } catch (e) {
    isAdmin = false;
  }
  Navigator.pushReplacement(
    context,
    MaterialPageRoute(builder: (_) => MainScreen(isAdmin: isAdmin)),
  );
} else {
  Navigator.pushReplacement(
    context,
    MaterialPageRoute(builder: (_) => const WelcomeScreen()),
  );
}
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Container(
      width: double.infinity,
      decoration: const BoxDecoration(
        gradient: LinearGradient(
          begin: Alignment.topLeft,
          end: Alignment.bottomRight,
          colors: [Color(0xFF070B34), Color(0xFF111B5A), Color(0xFF070B34)],
        ),
      ),
    child: Stack(
      children: [
        Positioned(
          top: 80,
          right: 40,
          child: Container(
            width: 80,
            height: 80,
            decoration: BoxDecoration(
              shape: BoxShape.circle,
              color: Colors.purple.withOpacity(0.4),
            ),
          ),
        ),
      ],
    ),
  );
}

```

```
),
Positioned(
  bottom: -80,
  left: -50,
  child: Container(
    width: 250,
    height: 250,
    decoration: BoxDecoration(
      shape: BoxShape.circle,
      color: Colors.purple.withOpacity(0.25),
    ),
  ),
),
Center(
  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      Image.asset(
        "assets/images/memora_logo.png",
        width: 140,
        height: 140,
      ),
      const SizedBox(height: 25),
      Text(
        "MEMORA",
        style: GoogleFonts.poppins(
          color: Colors.white,
          fontSize: 50,
          fontWeight: FontWeight.bold,
        ),
      ),
      const SizedBox(height: 10),
      RichText(
        text: TextSpan(
          style: GoogleFonts.poppins(fontSize: 20),
          children: [
            const TextSpan(
              text: "Learn ",
              style: TextStyle(color: Colors.white),
            ),
          ],
        ),
      ),
    ],
  ),
),
```

```

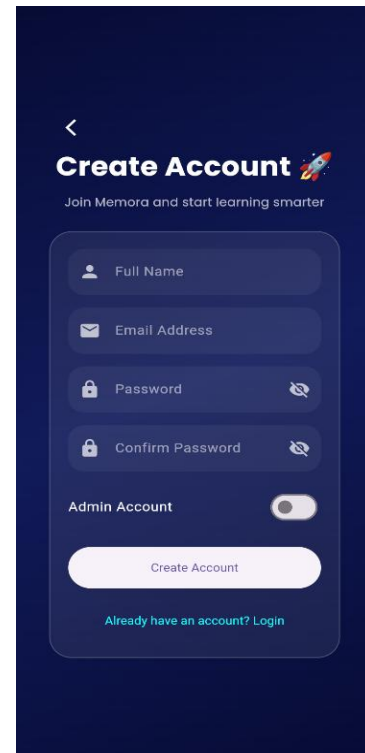
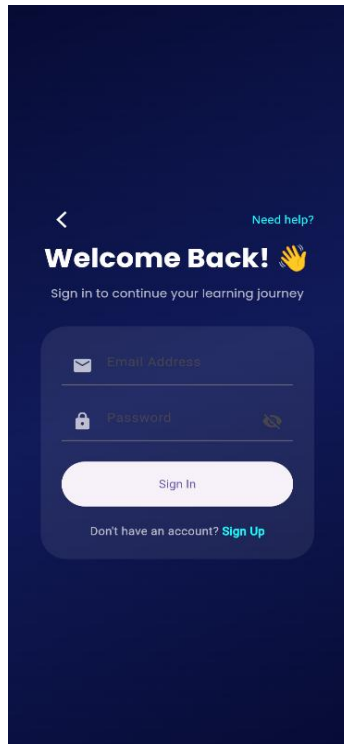
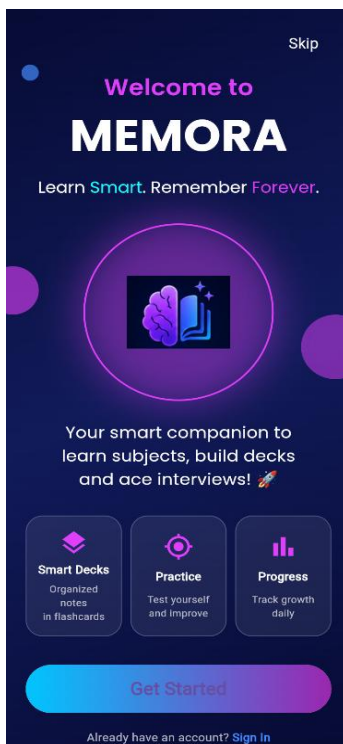
const TextSpan(
  text: "Smart",
  style: TextStyle(color: Colors.cyanAccent),
),
const TextSpan(
  text: ". Remember ",
  style: TextStyle(color: Colors.white),
),
const TextSpan(
  text: "Forever",
  style: TextStyle(color: Colors.purpleAccent),
),
],
),
),
const SizedBox(height: 50),
SizedBox(
  width: 250,
  child: LinearProgressIndicator(
    minHeight: 8,
    borderRadius: BorderRadius.circular(20),
  ),
),
const SizedBox(height: 20),
const Text(
  "Loading your knowledge...",
  style: TextStyle(color: Colors.white70),
),
],
),
],
),
),
);
}
}

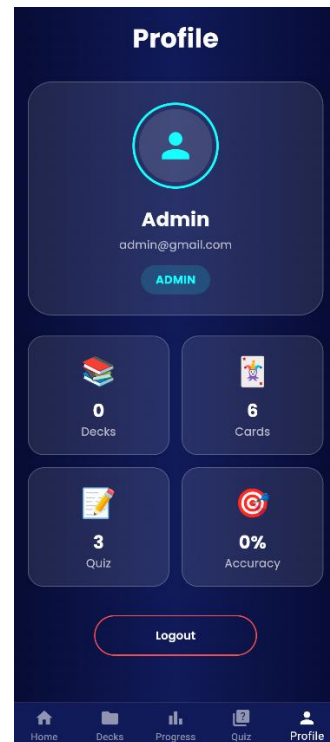
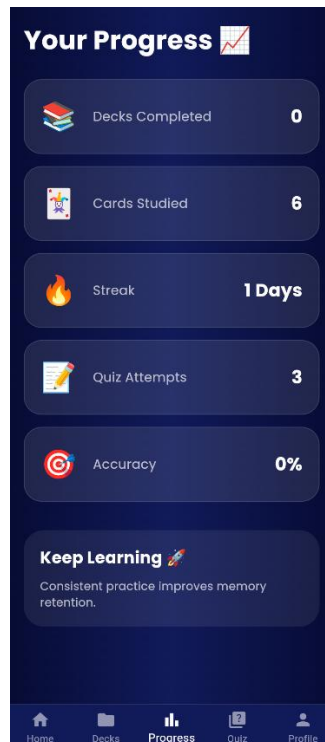
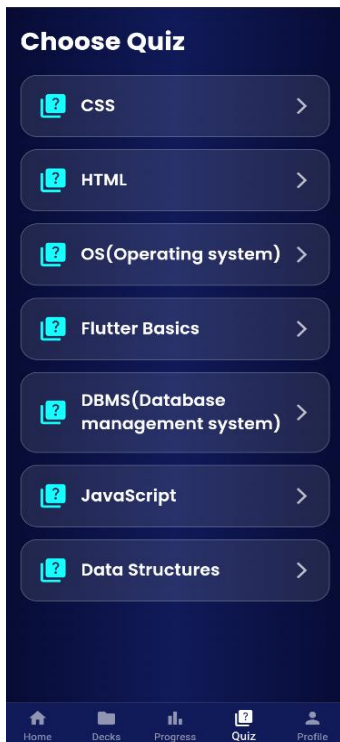
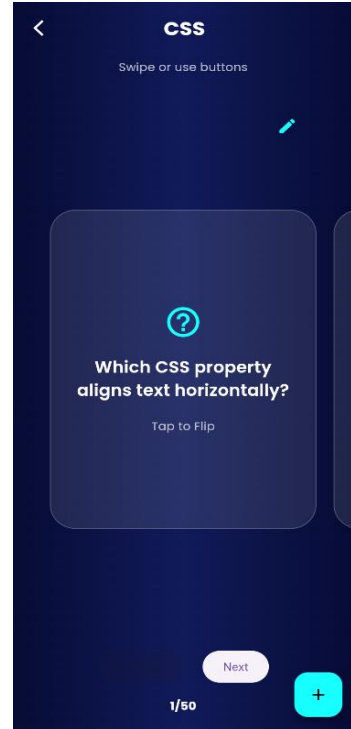
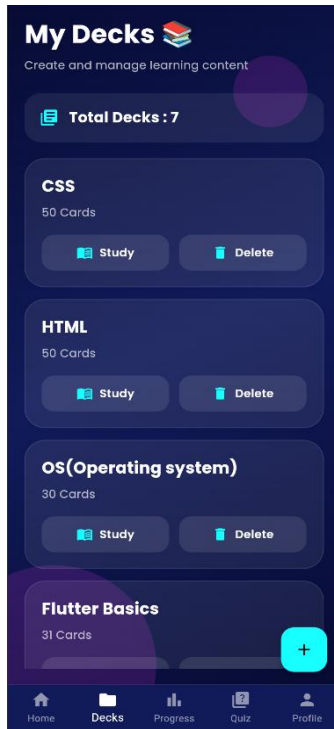
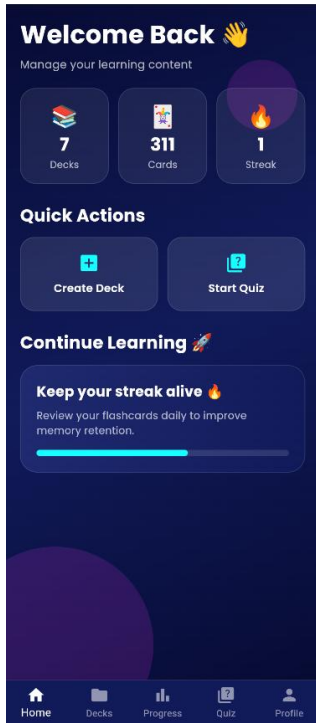
```


Screenshots:

The Memora application was successfully developed and tested using Flutter, Firebase Authentication, Cloud Firestore, and Provider state management. The application achieved all the objectives defined during the project planning phase and demonstrated reliable performance during functional testing. The integration of cloud-based services enabled secure data storage, real-time synchronization, and seamless user authentication.

The developed system provides a complete digital learning platform where users can create flashcard decks, add study materials, participate in quizzes, and monitor learning progress. The application offers a responsive user interface and smooth navigation across different modules, ensuring a positive user experience.





CONCLUSION:

Memora is a smart and user-friendly flashcards learning application developed using Flutter and Firebase to enhance the learning experience through digital technology. The application successfully integrates flashcard-based learning, quiz assessments, secure authentication, cloud data synchronization, and progress tracking within a single platform. By promoting active recall and self-paced learning, Memora helps users improve knowledge retention and monitor their academic performance effectively. The use of Flutter ensures cross-platform compatibility, while Firebase provides reliable and secure backend services. Overall, Memora offers an efficient, accessible, and scalable solution for students and lifelong learners, making the learning process more engaging, organized, and productive.

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